

Colegio de Montalban - Integrated Library Management System

Desktop Application: Comprehensive Features & Functionalities Guide

[!NOTE] This document provides a complete technical and functional breakdown of all modules, workflows, user interfaces, and capabilities available in the **Colegio de Montalban (CDM) Integrated Library Management System Desktop Application**.

Table of Contents

1. [Authentication & Security System](#)
 2. [Dashboard Module](#)
 3. [Book Catalog & Resource Management](#)
 4. [Student Directory & Patron Management](#)
 5. [Circulation: Book Borrowing & QR Scanner](#)
 6. [Online Reservations & Request Queue](#)
 7. [Circulation: Returns & Penalty Processing](#)
 8. [Reports, Analytics & Data Exporting](#)
 9. [Librarian Staff Management](#)
 10. [Institutional Guidelines & Terms Module](#)
 11. [System Architecture & Core Highlights](#)
-

1. Authentication & Security System

The desktop app features an enterprise-grade authentication system designed to restrict library administration access exclusively to verified library staff and head administrators.

Key Features:

- **Role-Based Access Control (RBAC):**
 - **Head Librarian / Admin:** Full administrative control over catalog, staff management, fine configurations, and database operations.
 - **Staff Librarian:** Access to daily circulation, book cataloging, student directory, returns processing, and reservations queue.
- **Librarian Sign-In Screen:**
 - Input fields for Username/Email and Password.
 - Show/Hide password toggle.
 - Demo login quick-fill shortcuts for rapid administrative testing (Admin Head Librarian , Staff Librarian).
- **Librarian Account Registration / Onboarding:**
 - Multi-step registration form requiring First Name, Last Name, Email, Phone Number, Official Employee ID, Role selection (Librarian vs Head Librarian), Username, and Password.
- **Production-Grade Account Recovery Modal:**
 - **Email OTP Reset (Recommended):** 3-step wizard (Enter Email → Verify 6-digit OTP → Set New Password) with live progress bar indicator and demo OTP fallback for offline development.

- **Employee ID Verification (Fallback):** Instant password reset using registered Username/Email and Official Employee ID (e.g. EMP-2024-001).
 - **JWT Session Management:**
 - **Automatic Authorization:** Bearer <token> interceptor attached to all client-side fetch requests for secure API communication.
-

2. 📊 Dashboard Module (DashboardPage)

The main command center for librarians, presenting real-time operational metrics, overdue warnings, quick actions, and automated synchronization.

Key Features:

- **Real-Time Key Metrics (Stats Grid):**
 - **Total Catalog Books:** Live count of physical and digital titles.
 - **Active Borrowed Books:** Number of books currently out on loan.
 - **Overdue Books Counter:** Real-time tally of past-due items requiring collection.
 - **Pending Online Reservations:** Real-time count of mobile app pickup requests waiting for librarian approval.
 - **Registered Students:** Total student patron directory count.
 - **Quick Actions Bar:**
 - Single-click navigation to *Scan QR / Borrow Book, Process Return, View Reservations, Add New Book, Register Student, and Generate Reports.*
 - **Overdue Notice & Penalty Alert Center:**
 - Highlights past-due transactions with student names, contact details, book titles, due dates, and auto-calculated overdue fines (₱10.00/day default rate).
 - Direct action buttons to process overdue returns or issue reminders.
 - **Recent Activity Stream:**
 - Chronological log of recent borrowings, returns, catalog updates, and reservation approvals.
 - **Automated Real-Time Polling:**
 - Background polling every **3 seconds** and window focus event listeners to instantly synchronize mobile app reservations and borrow requests without requiring manual page refreshes.
-

3. 📖 Book Catalog & Resource Management (CatalogPage)

A complete cataloging suite with advanced searching, filter matrices, MARC 21 record formatting, and built-in PDF eBook reading.

Key Features:

- **Advanced Multi-Filter & Search Matrix:**
 - **Global Search:** Instant search by Title, Author, ISBN, Call Number, or Subject Keywords.
 - **Institute Filter:** Filter resources by academic institute:
 - ICS – Institute of Computer Studies
 - IE – Institute of Education
 - IAAS – Institute of Arts, Sciences & Teacher Education

- IB – Institute of Business
 - **Academic Filters:** Filter by Year Level (1st Year to 4th Year), Semester (1st Sem , 2nd Sem), and Category/Genre (e.g., Computer Science, Literature, Engineering, Mathematics).
 - **Availability & Sorting:** Filter by availability (*All, Available, Borrowed*) and sort by *Title, Borrow Volume, or Newest Addition*.
 - **Full Book Management (CRUD):**
 - **Add Book Modal:** Register new acquisitions with Title, Author, ISBN, Call Number, Category, Institute, Course, Year Level, Semester, Publisher, Year Published, Stock Copies, Shelf Location, Cover Image URL, and PDF file upload.
 - **Edit Book Modal:** Update metadata, cover artwork, or replace digital eBook files.
 - **Delete Book Modal:** Safe deletion prompt verifying dependencies before record removal.
 - **Book Detail, MARC 21 & Electronic Reader Modal (BookPreviewModal):**
 - **Tab 1: Overview & Metadata:** Cover art preview, physical shelf location, availability badges, borrowing statistics, and instant *Borrow Book* CTA.
 - **Tab 2: MARC 21 Bibliographic Tag Format:** Automatically formatted standard MARC 21 tags (001 , 020 ISBN , 082 Dewey Call Number , 100 Author , 245 Title , 260 Publication , 300 Physical Details , 650 Subject) with a **One-Click Copy MARC Record** button.
 - **Tab 3: Digital Reader / PDF Viewer:** Embedded PDF document viewer allowing librarians and patrons to read electronic books directly inside the desktop interface.
-

4. Student Directory & Patron Management (StudentsPage)

Manages student patron profiles, faculty exemptions, and account statuses.

Key Features:

- **Student Directory Table:**
 - Displays Student ID Number, Full Name, Email Address, Course (BSIT , BSED , BSBA , BEED , etc.), Year Level, Section, Status (*Active, Pending Verification, Suspended*), and Active Loans Count.
 - **Search & Course Filtering:**
 - Instant search by Student ID or Student Name.
 - **Student Profile Management:**
 - **Add Student:** Register new student accounts with full personal and academic information.
 - **Edit Student:** Modify student details, contact email, or course information.
 - **Delete Student:** Remove student profiles from the active directory.
 - **Faculty & Staff Exemption Support:**
 - Capability to designate or exempt faculty members/teachers from strict student borrowing limits and QR code constraints.
-

5. Circulation: Book Borrowing & QR Scanner (BorrowPage)

An accelerated circulation desk workflow combining hardware/camera QR scanning with manual patron lookup.

Key Features:

- **Dual Patron Identification & QR Scanner:**
 - **Integrated Desktop Camera QR Scanner:** Uses device camera to scan digital QR passes presented on the CDM Mobile App by students or faculty. Instantly extracts Student/Faculty ID and pre-populates patron profile data.
 - **Manual Search & Auto-Complete Dropdown:** Search student directory by ID or Name with real-time suggestion dropdown.
 - **Inline Quick Student Registration:** Modal drawer to immediately register a new student without abandoning the active borrowing transaction.
 - **Transaction Setup & Checkout:**
 - Book selection grid/search with real-time stock availability verification.
 - Customizable Loan Period dropdown (e.g. 3 days, 7 days, 14 days).
 - Auto-calculated Due Date based on academic calendar settings.
 - Terms & Conditions acknowledgment checkbox.
 - Digital transaction receipt generation upon successful checkout.
-

6. Online Reservations & Request Queue (ReservationsPage)

Manages hold requests placed by students via the Mobile App.

Key Features:

- **Reservation Queue Table:**
 - Displays Reservation Code, Book Title, Student Name, Request Date, Pickup Deadline, and Status.
 - **Status Classification:**
 - **Pending** : New online hold request awaiting librarian action.
 - **Approved (Ready for Pickup)** : Book reserved at the circulation counter.
 - **Completed** : Book successfully collected by the student.
 - **Cancelled / Expired** : Reservation cancelled by user or expired past pickup window.
 - **Workflow Actions:**
 - **Approve Request:** Reserves physical copy and notifies student mobile app.
 - **Reject Request:** Declines reservation with optional reason.
 - **Convert to Active Borrow:** Instantly converts an approved reservation into an active loan upon physical pickup.
-

7. Circulation: Returns & Penalty Processing (ReturnsPage)

Processes returned books, logs item conditions, and manages fine collection.

Key Features:

- **Active Loan Registry & Search:**
 - Search active loans by Transaction ID, Student ID, Student Name, or Book Title.
- **QR / Barcode Rapid Return Scanner:**
 - Scan student QR pass or book barcode to instantly pull up active loan details.
- **Automated Overdue Penalty Calculator:**
 - Calculates exact overdue days and applies the institutional penalty rate (₱10.00/day).

- **Return Conditions & Payment Tracking:**
 - Log book condition (*Good Condition, Damaged, Lost Book*).
 - Record fine payment status (*Paid, Unpaid / Added to Student Account*).
 - Automatically returns book copy to available catalog stock upon completion.
-

8. Reports, Analytics & Data Exporting (ReportsPage)

Comprehensive reporting tools providing visual analytics and spreadsheet/PDF exports for institutional reporting.

Key Features:

- **Interactive Data Visualizations:**
 - **Circulation Trends:** Visual charts representing daily, weekly, and monthly borrowing activity.
 - **Institute Utilization Breakdown:** Bar graphs showing usage distribution across ICS , IE , IAAS , and IB .
 - **Top Borrowed Titles:** Leaderboard of most popular catalog resources.
 - **Summary KPI Cards:**
 - Total Circulation Volume, Total Fines Collected, Outstanding Unpaid Fines, and On-Time Return Compliance Rate.
 - **Exporting Capabilities:**
 - **CSV Spreadsheet Export:** Export catalog lists, active borrowing logs, overdue records, and transaction histories into standard .csv files.
 - **Print-Ready Reports:** Formatted print layout for generating physical PDF reports for library administration and department heads.
-

9. Librarian Staff Management (LibrariansPage)

Administrative directory for managing library personnel and access privileges.

Key Features:

- **Staff Roster Table:**
 - Displays Librarian Name, Employee ID, Email, Role (Head Librarian vs Staff Librarian), Registration Date, and Account Status.
 - **Administrative Controls:**
 - Head Librarians can review staff activity, update employee credentials, and adjust role authorizations.
-

10. Institutional Guidelines & Terms Module (TermsPage)

Central repository of official Colegio de Montalban library rules and operational guidelines.

Key Features:

- Detailed documentation covering:
 - Maximum borrowing quotas per patron category (Students vs Faculty).
 - Standard loan durations and renewal policies.

- Overdue penalty rates (₱10.00/day) and lost item replacement regulations.
- Library operating hours and digital eBook usage guidelines.

11. System Architecture & Core Highlights

Layer	Technology / Implementation
Frontend Framework	React 18 with TypeScript & Vite
Styling & Icons	Tailwind CSS with Lucide React SVG icons
Animation & Motion	Framer Motion (<code>motion/react</code>) for smooth page transitions and modal morphing
Backend API	Node.js & Express RESTful API server (<code>server/server.js</code>)
Database & Persistence	SQLite / LowDB JSON database with ACID transaction safety
Real-Time Sync	Auto-polling every 3s + window focus events for instant synchronization with Mobile App
Standard Integration	Full MARC 21 tag formatting & PDF reader embedded viewer
Authentication	JWT Bearer Tokens with Email OTP verification endpoint